

A Formulation Audit of Erdos Problem #728

Status: internal sandbox finding; not claimed as novel; not worth publishing as a standalone mathematical result.

Source page: <https://www.erdosproblems.com/728>

Summary

The literal written formulation of Erdos Problems #728 is true for a degenerate reason. It lower-bounds a and b , but does not upper-bound them. Thus one can take $a > n$ and make b very large so that the factorial divisibility is automatic.

This is not a solution of the intended logarithmic-barrier problem. It is only a statement-normalization check.

Written Statement

For sufficiently small $C > 0$ and $\epsilon > 0$, are there infinitely many integers a, b, n such that

$a \geq \epsilon n$, $b \geq \epsilon n$, $a! b!$ divides $n! (a+b-n)!$,
and
 $a+b > n + C \log n$?

Result

For every fixed $C > 0$ and every fixed $\epsilon > 0$, the written statement has infinitely many solutions.

Proof

Fix $C > 0$ and $\epsilon > 0$. For any sufficiently large positive integer n , choose an integer

$w \geq \max(C \log n, \epsilon n)$.

Define

$a = n + w + 1$, $b = (n+w+1)! / n! - 1$.

Then $a \geq \epsilon n$, and $b \geq \epsilon n$ for all sufficiently large n . Also $a+b-n = b+w+1$. Now compute

$$\frac{n! (a+b-n)!}{a! b!} = \frac{n! (b+w+1)!}{(n+w+1)! b!}$$

Since

$$(n+w+1)! / n! = b+1,$$

we get

$$\begin{aligned} \frac{n! (a+b-n)!}{a! b!} &= \frac{(b+1)(b+2)\dots(b+w+1)}{b+1} \\ &= (b+2)(b+3)\dots(b+w+1), \end{aligned}$$

which is an integer. Hence $a!b!$ divides $n!(a+b-n)!$. Finally, $a+b-n = b+w+1 > C \log n$, so $a+b > n + C \log n$. Since n can be chosen arbitrarily large, this gives infinitely many triples.

Interpretation

The construction uses $a > n$ and huge b . Therefore it does not touch the intended nontrivial problem. A meaningful formulation needs additional upper bounds, for example $a, b \leq (1-\delta)n$.

Even adding only $a, b \leq n$ would not fully remove degeneracy, since $a=b=n$ gives $n!n!$ dividing $n!n!$ and $2n > n + C \log n$ for all sufficiently large n .

Verdict

Useful as a formulation audit, but not a publishable new finding. The active search for publishable work should move to a genuinely open target or to a strengthened version of #728 with the intended hypotheses.